

The empirical recurrence for OEIS A251040: recovering a conjecture that is not written down

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Abstract

OEIS A251040 records “Empirical recurrence of order 60”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 176$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251040 is “Number of $(3+1)X(n+1)$ 0..3 arrays with nondecreasing $\min(x(i,j), x(i,j-1))$ in the i direction and nondecreasing absolute value of $x(i,j) - x(i-1,j)$ in the j direction.”. It has offset 1 and begins

3550, 45150, 422538, 3433899, 24844850, 165531421, 1034381160, 6148260175, ...

The entry states:

Empirical recurrence of order 60 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:02:08, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 176$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 176$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{60} c_i a(n-i),$$

c_1	19	c_{16}	−1054421803	c_{31}	413534683051	c_{46}	18241418831
c_2	−97	c_{17}	−5880986407	c_{32}	−1183595048499	c_{47}	−1029823844
c_3	−339	c_{18}	6160227582	c_{33}	−344091225169	c_{48}	−4271088755
c_4	4551	c_{19}	19616577061	c_{34}	1159102909767	c_{49}	286989664
c_5	−4640	c_{20}	−25419058004	c_{35}	228640885972	c_{50}	766337729
c_6	−82170	c_{21}	−53089968679	c_{36}	−945528568860	c_{51}	−55044068
c_7	226765	c_{22}	80625480502	c_{37}	−117002032373	c_{52}	−101740611
c_8	816626	c_{23}	117716862065	c_{38}	640486721607	c_{53}	7206931
c_9	−3707930	c_{24}	−204089094109	c_{39}	42241519328	c_{54}	9497214
c_{10}	−4594512	c_{25}	−215080296881	c_{40}	−358399299515	c_{55}	−610418
c_{11}	37205066	c_{26}	420726493376	c_{41}	−7498552964	c_{56}	−575884
c_{12}	8497342	c_{27}	324588951575	c_{42}	164427922323	c_{57}	29604
c_{13}	−263811685	c_{28}	−714786344790	c_{43}	−2217862903	c_{58}	19800
c_{14}	93308526	c_{29}	−404246126783	c_{44}	−61216294904	c_{59}	−600
c_{15}	1409237017	c_{30}	1007816557026	c_{45}	2379681565	c_{60}	−288

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60 + 4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

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